

**ENSURING ACCESS TO REGISTERED PHARMACEUTICAL PRODUCTS IN
LICENSED PHARMACEUTICAL PREMISES:**

The Role of Regulatory Agencies

By:

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Abstract

This paper discusses access to registered pharmaceutical products—including nanomedicines—in licensed pharmaceutical premises, and highlights the role of regulatory agencies, notably the **National Agency for Food and Drug Administration and Control (NAFDAC)** and the **Pharmacists Council of Nigeria (PCN)**.

A breakthrough in drug invention, whether for prophylaxis or curative purposes, is only one part of the success story in drug development. The processes of drug identification, extraction, evaluation, dosage form selection, and clinical trials that guarantee safety and efficacy do not automatically ensure that products are available where they are most needed—at the tail end of the distribution chain, namely retail and hospital pharmacies.

The paper concludes by proposing strategies to guarantee access to safe, effective, and genuine medicines—whether locally produced or imported—registered by NAFDAC, and distributed exclusively within registered, inspected, approved, and licensed pharmaceutical premises by the PCN.

Among the recommendations:

- **Manufacturers and importers** should utilize computerized invoicing systems to strengthen traceability of pharmaceutical products.
- **Retail and hospital pharmacies** should adopt patient-centered dispensing software that records patient profiles, enables product recalls, and facilitates communication between patients and pharmacists.

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1.0 Introduction

Pharmaceutical products prescribed, dispensed, or administered within healthcare institutions fall into two broad categories:

1. **Prescription drugs**
2. **Non-prescription (Over-the-Counter, OTC) drugs**

In Nigeria, there are two officially recognized sources of pharmaceutical products:

- **Local manufacturing**, and
- **Importation** from approved overseas pharmaceutical companies.

Correspondingly, there are also two primary outlets where patients or consumers access these products:

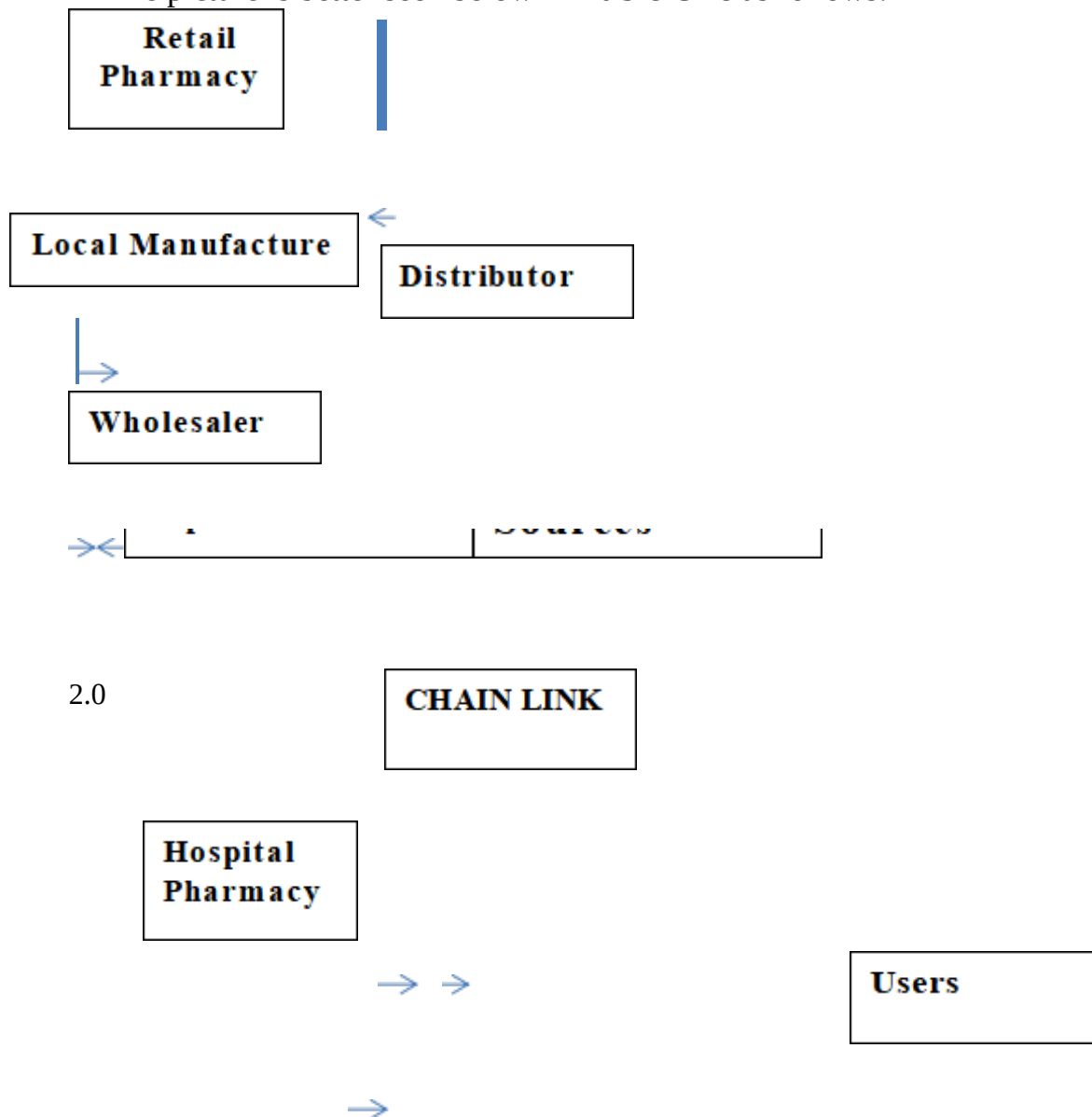
- **Retail pharmacies**, and
- **Hospital pharmacies**.

Both outlets function at the tail end of the drug distribution chain, sourcing products either from local manufacturers or from major importers.

Between these two ends of the chain are **distributors and wholesalers**, who serve as intermediaries. Positioned strategically, they often exert significant control over the pharmaceutical business in Nigeria.

The picture is better seen below in **Table One** as follows: -

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2.0 DIFFERENTIATING CORE PHARMACEUTICAL SCIENCE COURSES:

The Relevance of Pharmacy Management Courses

A plant is essentially made up of its root, stem, leaves, and fruits. A plant with medicinal value may contain the active pharmaceutical ingredient in any of these parts—or in all of them—in varying concentrations.

2.1 Pharmacognosy

Pharmacognosy is the branch of pharmacy education that focuses on the medicinal importance of plants. The process of separating the active pharmaceutical component from the root, stem, leaf, or fruit is known as **maceration**. Over time, various sophisticated techniques have been developed to separate different constituent groups in medicinal plants. Among these, **chromatography** is widely used, as it enables separation based on melting or boiling points of compounds, among other properties.

2.1.2 Pharmaceutical Chemistry

Once the active medicinal product has been identified, the next step is to isolate it in sufficient quantity. Often, very large amounts of raw plant material are required, which may be impractical. **Pharmaceutical Chemistry** enables us to identify the structure of the active ingredient using instruments such as the **ultraviolet (UV) spectrophotometer**.

Beyond identification, the structure can be synthesized in the laboratory, and the medicinal activity of the synthetic equivalent compared with the naturally extracted form. Chemical modifications of the identified structure can also be made to optimize efficacy, enhance safety, reduce toxicity, minimize side effects, improve compliance, and decrease resistance. Similarly, for inorganic chemicals of medicinal value, elements within the same group of the **Periodic Table** may be substituted until the most suitable compound for therapeutic use is achieved.

2.1.3 Pharmacology

Once the active ingredient has been isolated and characterized, the next stage is **pharmacological testing**. This involves studying its activity in living tissues.

Preclinical studies often begin with laboratory animals—mice, rats, rabbits, or monkeys—before progressing to **clinical trials** in human volunteers.

These studies determine the **therapeutic dose, lethal dose, side effects**, and the **optimum clinical dose**. The results form the basis for safe and effective therapeutic use.

2.1.4 Pharmaceuticals

After pharmacological validation, attention shifts to identifying the best formulation for stability, safety, and efficacy. This is the role of **Pharmaceutics**, which determines the most appropriate dosage form—whether tablets, capsules, creams, injectables, or inhalation.

Critical formulation considerations include:

- Stability of the product,
- Expiry date of the active ingredient,
- Compatibility with additives such as coloring agents and preservatives,
- Dissolution and disintegration times for solid dosage forms, and
- Resistance to breakage (friability).

These parameters ensure the delivery of a high-quality pharmaceutical product.

2.1.5 Clinical Pharmacy and Pharmacy Management

With the product formulated, **Clinical Pharmacy** evaluates its effects in live patients in hospital settings. Interactions with clinicians, pharmacists, and nurses are studied, including **drug-drug, food-drug, and drug-diagnostic reagent** interactions.

However, before clinical use is possible, **Pharmacy Management** becomes critical. It ensures that medicines reach patients efficiently through organized systems of importation, manufacturing, distribution, and retail. Pharmacy Management involves the coordinated use of scarce resources—**man, money, materials, machines, and methods**—across pharmaceutical industries, wholesalers, distributors, hospital pharmacies, and retail pharmacies.

Poor or ineffective management can result in unavailability or unaffordability of medicines, regardless of how well they have been researched or developed.

In essence, **Pharmacy Management is to pharmacy what Community Medicine is to medicine**: a discipline that ensures products reach patients in a safe, affordable, and professional manner.

3.0 The Roles of Drug Regulatory Agencies

Two major regulatory agencies of the Federal Government oversee drug matters in Nigeria:

1. **The National Agency for Food and Drug Administration and Control (NAFDAC)** – Established in 1993 by Decree (now Act) 13, NAFDAC regulates and controls the importation, exportation, manufacture, advertisement, distribution, sale, and use of food, drugs, cosmetics, medical devices, bottled water, and chemicals. It is responsible for the **registration of all pharmaceutical products** and ensures compliance with **current Good Manufacturing Practice (cGMP)**.
2. **The Pharmacists Council of Nigeria (PCN)** – Established in 1992 by Decree (now Act) 91, the PCN regulates and controls pharmacy education and practice in all its dimensions. It inspects, approves, registers, licenses, and monitors all pharmaceutical premises, including manufacturing plants, importation firms, wholesale outlets, retail pharmacies, hospital pharmacies, and patent medicine shops.

The relationship between NAFDAC and PCN is **symbiotic**:

- Only **NAFDAC-registered products** should be stocked and dispensed in **PCN-licensed premises**.
- Any product not registered by NAFDAC is **fake**, regardless of origin.
- Any outlet not licensed by PCN is **illegal**, regardless of location.

Together, NAFDAC and PCN play complementary roles in safeguarding the health of Nigerians.

4.0 Sources of Drugs in Nigeria

There are two official sources of drugs in Nigeria:

1. **Local manufacturing companies**, represented by the **Pharmaceutical Manufacturers Group of the Manufacturers Association of Nigeria (PMG-MAN)**.
2. **Licensed importers**, represented by the **Association of Pharmaceutical Importers of Nigeria (APIM)**.

Both bodies are central to any effective and transparent distribution model. Additionally, the **Federal Ministry of Health (FMOH)**, through its Food and Drugs Services Department, works in collaboration with NAFDAC and PCN to sanitize the distribution system.

If every drug produced locally or imported is **registered by NAFDAC** and **distributed only through PCN-licensed premises**, patients and consumers would be guaranteed access to safe and effective medicines.

5.0 The Situation with Nanomedicines

The current distribution system for orthodox pharmaceutical products in Nigeria is widely acknowledged as inadequate. The question arises: should this same flawed model be applied to emerging therapies such as **nanomedicines**? The answer is clearly **no**.

Advances in **DNA research, nanosciences, and nanotechnologies** promise innovative treatments. However, without a reliable distribution model, these breakthroughs could fail to reach patients safely.

As Molly Stevens and Clive Roberts note in *Pharmacy 2020*:

“...the increasing demands of understanding how modern medicines work at the molecular level, the shift towards predictive, preventive and personalized healthcare, and challenges from nanotechnology and stem cell technology have all added to the need for pharmacists to remain the experts in medicines.”

Ultimately, the responsibility of pharmacists is not only to understand medicines but also to ensure **patients have safe and timely access** to them.

6.0 Computer Software Solutions

A computerized monitoring system can revolutionize pharmaceutical distribution in Nigeria. Just as the **Central Bank of Nigeria (CBN)** monitors financial transactions nationwide through electronic systems, a similar approach can be adopted in pharmacy.

A pharmaceutically-designed software system would:

- Track suppliers, wholesalers, retailers, and patients across the distribution chain;
- Require importers and manufacturers to log all transactions at the point of invoicing;
- Ensure traceability of products from source to end-user;
- Operate only within **PCN-licensed premises**.

Such an electronically monitored regime would greatly enhance transparency, accountability, and patient safety.

7.0 Conclusion

This paper has reviewed the current distribution model for orthodox pharmaceutical products in Nigeria, its limitations, and the urgent need for reform in preparation for future innovations such as nanomedicines.

By adopting **computerized monitoring systems** and strengthening collaboration between **NAFDAC, PCN, PMG-MAN, and APIM**, Nigeria can establish a robust, transparent, and patient-centered distribution model.

Ultimately, the goal is to guarantee that all medicines—whether conventional or nanomedicines—are safe, efficacious, and accessible to the patients who need them most.